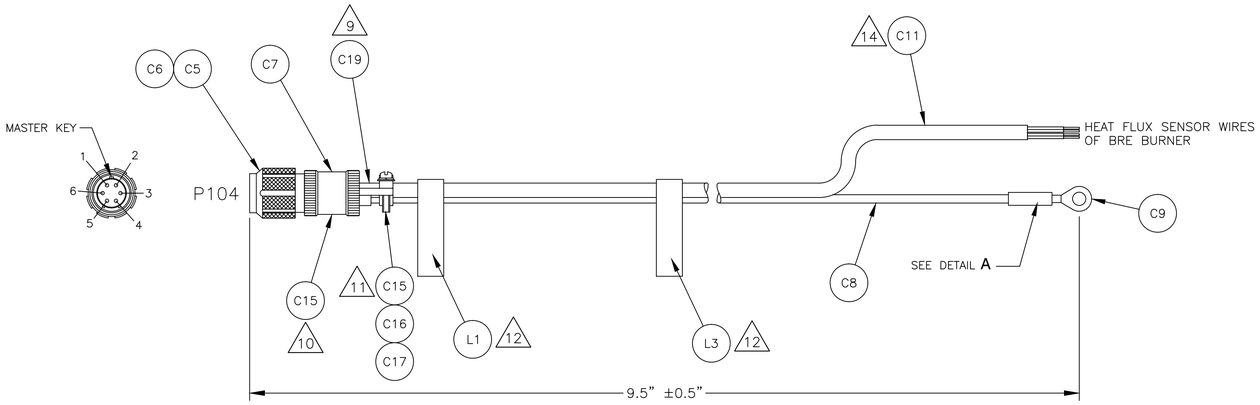
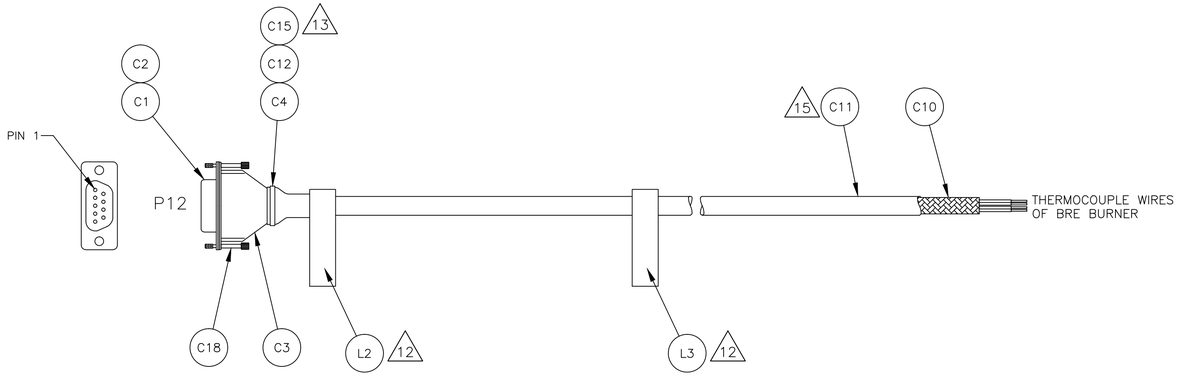
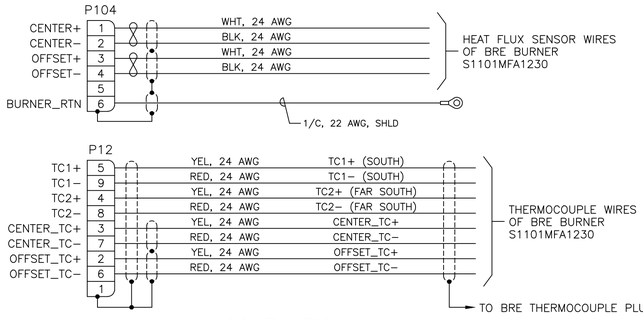
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION	MANUFACTURER
C1	1	SND9S1000G	CONNECTOR, D-SUB, FEMALE, 9 POS	POSITRONIC
C2	9	FC6020M2	CONTACT, SOCKET, SIZE 20, AWG 20-24	POSITRONIC
C3	1	557-398UC1B	BACKSHELL, D-SUB, SIZE 1	GLENAIR
C4	A/R	600-057	BACKSHELL MICRO BAND	GLENAIR
C5	1	805-002-16UC8-6PA	CONNECTOR, CIRCULAR, 6 POS.	GLENAIR
C6	6	809-001	CONTACT, PIN, SIZE 23, AWG 22-26	GLENAIR
C7	1	380MS135UC1104-6	BACKSHELL, STRAIGHT, SIZE 11	GLENAIR
C8	A/R	M27500-22RC1S06	1/C, 22 AWG, SHLD	QPL
C9	1	321884	TERMINAL RING, #4, 22-16 AWG	TE CONN.
C10	A/R	2142SV005	BRAIDED EMI SHIELDING, 1/4"	ALPHA
C11	A/R	5011000632S	EXPANDABLE SLEEVING, 1/4"	BENTLEY HARRIS
C12	1	PLT1M-M702Y	CABLE TIE, 4", FLUOROPOLYMER	PANDUIT
C13	A/R	THFE4-115009	SHRINK TUBING, EXPANDED PREHEAT ID .115"	SAINT GOBAIN
C14	A/R	THFE4-180010	SHRINK TUBING, EXPANDED PREHEAT ID .180"	SAINT GOBAIN
C15	A/R	2216 B/A GRAY	EPOXY, 2 PART	3M
C16	A/R	7649	PRIMER N	LOCTITE
C17	A/R	222MS	THREADLOCKER	LOCTITE
C18	2	687-316EMS2	1" JACKSCREW, KNURLED, SOCKET HEAD	GLENAIR
C19	A/R	THFE4-310011	SHRINK TUBING, EXPANDED PREHEAT ID .310"	SAINT GOBAIN

LABEL LIST	
ITEM	INSCRIPTION
L1	P104
L2	P12
L3	S1101EFA1244 SN xxxx







SCHEMATIC

16. HARNESS TO BE BUILT IN CONJUNCTION WITH CORRESPONDING BRE BURNER ASSEMBLY S1101MFA1230.

17. PLACE COMPLETED ASSEMBLY IN CLEAN, STATIC-SHIELDING PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/STATEMENT OF WORK PER SAE AS478, METHOD 35D. STORE IN A PROJECT APPROVED STORAGE AREA.

NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.3, NASA-STD-8739.4, AND FOLLOWING NOTES:
   a. ALL SHIELD TERMINATIONS ARE INTERFERENCE SENSITIVE.
5. FOR TENSILE PULLS SEE TENSILE PULL STRENGTH LIST. TENSILE PULL STRENGTHS PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS.
6. INSTALL CONTACTS IN ALL UNUSED POSITIONS OF CIRCULAR AND D-SUB CONNECTORS.
7. PERFORM CONTACT RETENTION TEST PER NASA-STD-8739.4 AND/OR MANUFACTURER'S SPECIFICATIONS. CONTACTS REQUIRING A PULL TEST FOR RETENTION SHALL BE PERFORMED AS FOLLOWS: A GENTLE PULL ON THE WIRE SHALL BE USED TO DETERMINE IF THE CONTACT IS SEATED PROPERLY.
8. TERMINATE SHIELDS INTO BACKSHELL PER MANUFACTURER'S SPECIFICATION.

△9 TO SAFELY SECURE CABLE IN STRAIN RELIEF INCREASE THE CABLE DIAMETER USING HEAT SHRINK TUBING (ITEM C19). MAXIMUM OF TWO (2) LAYERS.

△10 TORQUE BACKSHELL (ITEM C7) TO 14-18 IN-LBS. STAKE BACKSHELL USING EPOXY (ITEM C15).

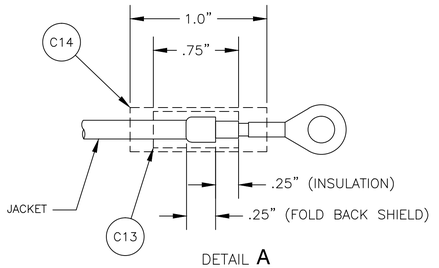
△11 TRIM FASTENERS ON STRAIN RELIEF TO EXPOSE 1-2 THREADS. APPLY PRIMER N (ITEM C16) AND THREADLOCKER 222MS (ITEM C17), TORQUE FASTENERS PER CHART. APPLY EPOXY (ITEM C15) TO EXPOSED THREADS.

△12 APPLY TEMPORARY CABLE IDENTIFICATION LABELS P/N TZE-211.
NOTE: TO BE REMOVED AT INSTALLATION INTO FINAL ASSEMBLY.

△13 APPLY EPOXY (ITEM C15) TO SHARP EDGE OF MICRO BAND (ITEM C4).

△14 LENGTH OF HARNESS FROM FACE END OF P104 TO TAB TIP OF BRE PLUG ENTRY STRAIN RELIEF TAB - (SEE NEXT ASSEMBLY S1101MFA1230) - TBD

△15 LENGTH OF HARNESS FROM FACE END OF P12 TO TAB TIP OF BRE PLUG ENTRY STRAIN RELIEF TAB - (SEE NEXT ASSEMBLY S1101MFA1230) - TBD



DETAIL A

CRIMP TOOL LIST			
CONTACT P/N	CRIMP TOOL	POSITIONER	I/E TOOL
809-001	M22520/2-01	K1461	809-088
FC6020M2	M22520/2-01	M22520/2-08	M81969/1-02
321884	46447 OR EQUIV.	N/A	N/A

TENSILE PULL STRENGTH LIST				
MANUFACTURER	CONTACT P/N	WIRE AWG	LBS	
GLENAIR	809-001	22	8.0	(NICKEL)
		22	12.0	(SILVER)
		24	6.0	
POSITRONIC	FC6020M2	24	8.0	
TE CONN.	321884	22	15.0	

BACKSHELL CLAMPING TORQUE TABLE		
SCREW SIZE	TORQUE RANGE	UNITS
2-56	2 ±0.5	IN-LB
4-40	4 ±0.5	IN-LB
6-32	6 ±1.0	IN-LB
8-32	8 ±1.0	IN-LB
10-32	10 ±1.0	IN-LB
.250-20	12 ±1.0	IN-LB

REV	DESCRIPTION	APPROVED
A	REVISED PER ACME-ECO-482.	MH 11/23/15 TG 11/24/15 DS 11/24/15

REVISION HISTORY

S1101MFA1230
NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E. McCLOSKEY 11/25/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 7/7/14	QUALITY	D. SEELEY 7/14/14
REVIEWER	J. BRUEWER 7/7/14	MATERIALS	M. WILKINSON 7/7/14
MECHANICAL	A. DRAKE 7/11/14	PROJ LEAD	B. BOROWSKI 7/21/14
ELECTRICAL	T. COBELI 7/8/14		
STRUCTURAL	C. WERNER 7/9/14		
SAFETY	W. BORRELLI 7/7/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: S2-BT3

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
BRE 50mm
HARNESS ASSEMBLY

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EFA1244	A

SCALE NONE | INITIAL REL DATE 07/21/14 | SHEET 1 OF 1

RELEASE LEVEL: RELEASED
MODEL NAME: S1101EFA1244_REV_A.DWG

AutoCAD 2012

DWG NO S1101EFA1244 | SH 1 | REV A